

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	5 July 2026
Team ID	
Project Name	Book A Doctor
Maximum Marks	4 Marks

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application (Web UI)	HTML, CSS, JavaScript, React.js (Vite)
2.	Application Logic	Backend routing, controllers, and business logic	Node.js, Express.js
3.	Database	Data Type, Configurations etc.	MongoDB (Mongoose ODM)
4.	Authentication	Securing user sessions	JWT (JSON Web Tokens), bcrypt
5.	File Storage	File storage requirements (If profile pics are added)	Local Filesystem (uploads folder)
6.	Infrastructure	Application Deployment	Local Server (Node), easily deployable to cloud (e.g., Render, Vercel, Heroku)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	React.js, Express.js, Node.js
2.	Security Implementations	List all the security / access controls implemented	JWT for route protection, bcrypt for password hashing, Express middleware
3.	Scalable Architecture	Justify the scalability of architecture	3-tier architecture (Client, Server, Database) allows independent scaling of frontend and backend.
4.	Availability	Justify the availability of application	Can be deployed to highly available cloud services like AWS or MongoDB Atlas.
5.	Performance	Design consideration for the performance	Vite for fast frontend building, React for efficient DOM updates, asynchronous Node.js operations.